Exp No:17 Experiment to understand Linear Regression for a given

data set.

AIM:

To implement a linear regression model for predicting marks on study hours.

Description:

Understand the Linear regression for the dataset given.

CODE:

```
import numpy as np
import pandas as pd
df=pd.read_csv('Salary_data.csv')
df
df.info()
df.dropna(inplace=True)
df.info()
df.describe()
features=df.iloc[:,[0]].values
label=df.iloc[:,[1]].values
from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test=train_test_split(features,label,test_size=0.2,random_st
from sklearn.linear_model import LinearRegression
model=LinearRegression()
model.fit(x_train,y_train)
model.score(x_train,y_train)
model.score(x_test,y_test)
model.coef_
model.intercept_
import pickle
pickle.dump(model,open('SalaryPred.model','wb'))
```

```
model=pickle.load(open('SalaryPred.model','rb'))

yr_of_exp=float(input("Enter Years of Experience: "))

yr_of_exp_NP=np.array([[yr_of_exp]])

Salary=model.predict(yr_of_exp_NP)

print("Estimated Salary for {} years of experience is {}: " .format(yr_of_exp,Salary)
```

OUTPUT:

1. Original Dataset

```
   StudyHours  Marks
0          1    35
1          2    40
2          3    45
3          4    50
4          5    55
5          6    60
6          7    65
7          8    70
8          9    74
9         10    79
10        11    83
11        12    87
12        13    91
13        14    94
14        15    97
```

2. Dataset Information

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 15 entries, 0 to 14
Data columns (total 2 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   StudyHours  15 non-null     int64
 1   Marks       15 non-null     int64
dtypes: int64(2)
memory usage: 372.0 bytes
```

3. Statistical Summary

```
       StudyHours     Marks
count   15.000000  15.000000
```

mean	8.000000	68.333333
std	4.472136	20.282528
min	1.000000	35.000000
25%	4.500000	52.500000
50%	8.000000	70.000000
75%	11.500000	85.000000
max	15.000000	97.000000

4. Linear Regression Model

LinearRegression()

5. Training Score

0.995281820510348

6. Testing Score

0.9961244386254701

7. Model Coefficient

array([[4.51970701]])

8. Model Intercept

array([32.0523195])

9. Final Prediction

Enter Study Hours: 4
Estimated Marks for 4.0 hours of study is [[50.13114754]]

RESULT:

The linear regression model was successfully created and used to predict marks based on the number of study hours.